Exercise path

Start with exercises 1 and 2. Then choose exercise 3 or 4 without help. Exercises 5 to 7 are a bank of transfer, investigation or programming tasks: the teacher assigns one. Exercise 8 is revisited later.

Exercise 8.1 · A distribution to construct

A fair die with six faces numbered from 1 to 6 is rolled. The random variable X assigns to the outcome the remainder of its Euclidean division by 3. For example, a result divisible by 3 gives $X = 0$.

1. For each of the six faces, give the value taken by X.
2. Present the probability distribution of X in a table and verify that the sum of probabilities equals 1.
3. Calculate $P(X < 2)$, stating which values of X are selected.

Exercise 8.2 · Two variance calculations

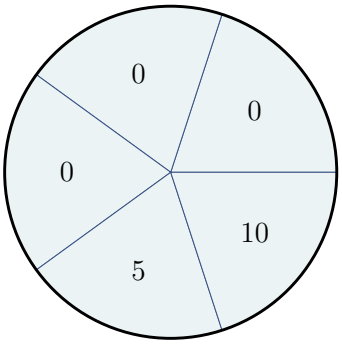
The random variable X takes the values -2, 0 and 4 with respective probabilities $\frac{1}{4}$, $\frac{1}{2}$ and $\frac{1}{4}$.

1. Calculate its expectation $\mu = E(X)$ and the moment $E(X^2)$.
2. Calculate $V(X)$ using the König-Huygens formula, then deduce the exact standard deviation.
3. Derive the variance again from the definition $V(X) = E((X-\mu)^2)$. Present the three contributions to the calculation.

Exercise 8.3 · A fair game

A wheel has five sectors of equal angle, bearing the numbers 0, 0, 0, 5 and 10. We assume the five sectors are equally likely. The number obtained represents a gross gain G, in monetary units. To play, you pay a stake m ; the net gain is therefore $X = G - m$.

1. Determine the distribution of G, then determine the stake m which makes the game fair, i.e. such that $E(X) = 0$.
2. After studying variance, calculate the standard deviation of the gross gain. Deduce that of the net gain when the stake is fair.
3. Explain why changing the stake changes the expected net gain but not its standard deviation.



The five sectors have the same angle. The three sectors showing 0 are distinct and must all be counted.

Exercise 8.4 · Same mean, different risk

Two fictional investments are offered. The first gives a certain gain $X = 1$. The second provides a gain $Y = -2$ with probability $\frac{1}{2}$ and a gain $Y = 4$ with probability $\frac{1}{2}$. The gains are expressed in the same currency unit.

1. Compare the expectations of the two gains, then their variances.

2. A person states: "These investments are equivalent since they have the same mean." Explain what this sentence leaves out and what the variance allows us to distinguish here.

Exercise 8.5 · Linearity without independence

A fair die numbered from 1 to 6 is rolled. Let X be the result of the roll and let $Y = 7 - X$. The two variables are therefore obtained from the same roll.

1. Calculate $E(X)$ and $E(Y)$, then $E(X + Y)$. Check the linearity of expectation on this example.

2. Are the variables X and Y independent? Justify using, for example, the events $\{X = 1\}$ and $\{Y = 6\}$.

3. Does the linearity of expectation require independence? Conclude based on your results.

Exercise 8.6 · A minimum loss

A random variable X takes finitely many values and has an expectation μ. For a real number a, we measure the mean squared error associated with the constant prediction a by $E((X - a)^2)$.

1. Prove that $E((X - a)^2) = V(X) + (a - \mu)^2$. You may write $X - a = (X - \mu) + (\mu - a)$.

2. Deduce the value of a which minimises this error, then the minimum value. Justify the uniqueness of the optimal choice.

Exercise 8.7 · Letter frequencies

We want to measure the frequency of the letter "a" in a text. The denominator is the number of alphabetic characters: spaces, digits and punctuation marks are excluded. Capital letters are treated as lower case; accented letters remain distinct from "a".

1. Write a Python function that returns this frequency. State the result chosen when the text contains no letters.
2. Test the function with `Aba !`, then with `123 !` and with the empty string. Explain the results expected before running the program.

Exercise 8.8 · Delayed review

We repeat twice, independently, an experiment whose probability of success is $\frac{1}{2}$. The random variable X counts the number of successes obtained.

1. Derive the distribution of X using the four possible sequences of outcomes. Do not use the general binomial distribution formula.
2. Calculate $E(X)$ and $V(X)$ starting from this distribution.

Independent work. Do only the task assigned at the end of the session, as set in the dated scheme of work: 10-20 min maximum. The exercises depend on the concepts already studied; some questions are therefore postponed. Revisit exercise 8 one week after the chapter (10 min). If the planned time is exceeded, note the difficulty and bring your attempts; do not try to do the whole reserve.

To be done in the CQFD textbook

These exercises are done in your textbook. The scheme of work gives the session; they replace the homework planned for that day (10 to 20 minutes in total).

CQFD, Exercise 28 p. 306, all questions. Calculate the expectation from the distribution table; keep the probabilities associated with each value.

CQFD, exercise 31 p. 306, all questions; use the distribution from exercise 28. Calculate the variance then the standard deviation, rounding to one decimal place as required by the textbook.